

Unidirectional Flushing (UDF) Plan – NAS II Sigonella

(*) Reference Section C, Annex 0200000 Management and Administration, Spec Item 2.6.2 Dissemination of Information

This plan outlines the work required to perform the “Unidirectional flushing” at NAS II. This results in water flowing toward the hydrant in only one direction resulting in higher flow velocity, more scouring and better cleaning of the pipes with less water use than the “Conventional flushing” This plan will be implemented once per year.

Safety procedure.

Apply before and during the UDF.

1. Necessary personal protective equipment and the correct tools:
 - a. Personal Protective Equipment (PPE)
 - i. Safety shoes
 - ii. Hard hat
 - iii. Hearing protection
 - iv. Safety glasses
 - v. Protective gloves
 - vi. Reflective vest (CAT 2)
 - b. Site Safety Equipment Required
 - i. Availability of communications (telephone or radio)
2. Employees tasked to complete flushing tasks need be minimum two-person due to the distances between hydrants and valves.
3. At least one crew member should be designated to control traffic while the other crew members are operating the distribution system valves.
4. Traffic signs, cones, flags, and/or vehicles with warning lights should be used as necessary to alert motorists of flushing activities.
5. Brightly colored safety vests shall be worn by all crew members.
6. Keep the public away from flushing activities. **Children are often attracted to such activities and should be kept away.**
7. Open hydrant valves completely to prevent water from discharging through the barrel drain at pressure.
8. Open and close hydrants slowly to prevent the development of dangerous pressure surges and water-hammer.
9. Use well-restrained energy dissipaters designed to prevent the water pouring out of the valve from damaging government and private property.
10. Discharge flushing water directly into a sanitary or storm sewer whenever possible to avoid flooding streets and underground electrical vaults. If street flooding is unavoidable, use signs, flags, and other warning devices to direct traffic away from the area. Notify Security and Fire Department about the UDF situation.

ATTACHMENT -1606001-07
UNIDIRECTIONAL FLUSHING PLAN NAS II

The desired minimum velocity is 6 feet per second. A single 2-1/2 inch outlet should be sufficient for the 6-inch and 8-inch mains. For larger mains, it may be more advantageous to open additional hydrants and open the 4-1/2 inch nozzle.

Utilize pitot gauges with flow diffusers to minimize grounds damage.

Main Size	Suggested Minimum Flow
(in)	(gpm)
6	550
8	950
10	1500
12	2100
14	2800

Note: In most cases, customers will not notice any changes to their water delivery during the UDF process, but UEM staff must inform the Fire Department and the sensitive customer (Hospital, school etc.) before to start the UDF.

Flushing Strategy: Loops 1 - 4 are intended to be conducted in one continuous operation. As such, (12) valves will be secured during Loop 1 and will remain closed until the conclusion of Loop 4.

Loop 1: (*) (Total of 12 Valves to Be Secured)

Site Plan "A"

Valve	Size	Location
	(mm)	
W126B	225	(*)
W260B	200	
W349B	315	
W671B	150	
W51B	200	
W52B	200	
W75B	150	

Site Plan "B"

Valve	Size	Location
	(mm)	
W674B	200	(*)
W138B	200	

ATTACHMENT -1606001-07
UNIDIRECTIONAL FLUSHING PLAN NAS II

Site Plan “C”

Valve	Size	Location
	(mm)	
W309B	200	(*)
W653B	355	
W78B	225	

Listed below is the sequence of hydrants to operate. Secure each hydrant before proceeding to the next location. Flow each hydrant until the flowing water clears. Record the approximate minutes of flushing performed at each location.

Sequence	Hyd ID	Location
1	FH94B	(*)
2	FH52B	
3	FH47B	
4	FH91B	
5	FH89B	
6	FH88B	

At the conclusion of Loop 1, do not reopen any valves as they need to remain secured for Loops 3 and 4.

Loop 2: (*)

Valves to Be Secured:

Valve	Size	Location
	(mm)	
W53B	150	(*)
W656B	150	

Listed is the sequence of hydrants to operate. Secure the noted hydrant before proceeding to the next location. Flow each hydrant the flowing water clears. Record the approximate minutes of flushing performed at each location.

Sequence	Hyd ID	Location
1	FH160B	(*)
2	FH162B	
3	FH172B	
4	FH175B	
5	FH170B	

ATTACHMENT -1606001-07
UNIDIRECTIONAL FLUSHING PLAN NAS II

Valves to Be Reopened:

Valve	Size	Location
	(mm)	
W53B	150	(*)
W656B	150	

Unlike Loop 1, reopen the (2) secured Loop 2 valves before proceeding to Loop 3.

Loop 3: (*)

Valves to Be Secured:

Valve	Size	Location
	(mm)	
W309B	150	(*)

Listed is the sequence of hydrants to operate. Secure the noted hydrant before proceeding to the next location. Flow each hydrant the flowing water clears. Record the approximate minutes of flushing performed at each location.

Sequence	Hyd ID	Location
1	FH85B	(*)
2	FH66B	
3	FH61B	
4	FH63B	

Valves to Be Reopened:

Valve	Size	Location
	(mm)	
W309B	150	(*)

Unlike Loop 1, reopen the secured Loop 3 valve before proceeding to Loop 4.

ATTACHMENT -1606001-07
UNIDIRECTIONAL FLUSHING PLAN NAS II

Loop 4: (*)

Valves to Be Secured:

Valve	Size	Location
	(mm)	
W310B	200	(*)
W658B	355	

Listed is the sequence of hydrants to operate. Secure the noted hydrant before proceeding to the next location. Flow each hydrant the flowing water clears. Record the approximate minutes of flushing performed at each location.

Sequence	Hyd ID	Location
1	FH48B	(*)
2	FH93B	
3	FH65B	
4	FH55B	
5	FH50B	
6	FH51B	
7	FH58B	

Loop #4 Valves to Be Reopened:

Valve	Size	Location
	(mm)	
W310B	200	(*)
W658B	355	

At the conclusion of Loop 4 flushing, reopen the twelve (12) valves securing during Loop 1 (listed below):

Site Plan "A"

Valve	Size	Location
	(mm)	
W126B	225	(*)
W260B	200	
W349B	315	
W671B	150	
W51B	200	
W52B	200	
W75B	150	

ATTACHMENT -1606001-07
UNIDIRECTIONAL FLUSHING PLAN NAS II

Site Plan "B"

Valve	Size	Location
	(mm)	
W674B	200	(*)
W138B	200	

Site Plan "C"

Valve	Size	Location
	(mm)	
W309B	200	(*)
W653B	355	
W78B	225	

Loop 5: (*)

Valves to Be Secured:

Valve	Size	Location
	(mm)	
W658B	355	(*)
W64B	355	
W524B	355	
W103B	225	
W72B	315	
W110B	280	
W101B	180	

Listed is the sequence of hydrants to operate. Secure the noted hydrant before proceeding to the next location. Flow each hydrant the flowing water clears. Record the approximate minutes of flushing performed at each location.

Sequence	Hyd ID	Location
1	FH143B	(*)
2	FH70B	
3	FH141B	
4	FH145B	
5	FH156B	
6	FH150B	
7	FH177B	
8	FH155B	
9	FH153B	
10	FH152B	

ATTACHMENT -1606001-07
UNIDIRECTIONAL FLUSHING PLAN NAS II

Valves to be reopened:

Valve	Size	Location
	(mm)	
W658B	355	(*)
W64B	355	
W524B	355	
W103B	225	
W72B	315	
W110B	280	
W101B	180	

Loop 6: (*)

Flushing Strategy: Loops 6 and 7 are intended to be conducted in one continuous operation.

Valves to Be Secured:

Valve	Size	Location
	(mm)	
W51B	200	(*)
W36B	355	

Listed is the sequence of hydrants to operate. Secure the noted hydrant before proceeding to the next location. Flow each hydrant the flowing water clears. Record the approximate minutes of flushing performed at each location.

Sequence	Hyd ID	Location
1	FH79B	(*)

Valves to remain closed for Loop 7:

Valve	Size	Location
	(mm)	
W36B	355	(*)

Valves to open for Loop 7:

Valve	Size	Location
	(mm)	
W51B	200	(*)

ATTACHMENT -1606001-07
UNIDIRECTIONAL FLUSHING PLAN NAS II

Loop 7: (*)

Additional valves to be secured:

Valve	Size	Location
	(mm)	
W52B	200	(*)

Listed is the sequence of hydrants to operate. Secure the noted hydrant before proceeding to the next location. Flow each hydrant the flowing water clears. Record the approximate minutes of flushing performed at each location.

Sequence	Hyd ID	Location
1	FH76B	(*)

Valves to be reopened:

Valve	Size	Location
	(mm)	
W52B	200	(*)
W36B	200	

Note: If you proceed directly to Loop 8, Valve 36B can remain closed.

Loop 8: (*)

Valves to Be Secured:

Valve	Size	Location
	(mm)	
W36B	355	(*)
W15B	225	
W358B	200	
W20B	225	
W1B	355	

ATTACHMENT -1606001-07
UNIDIRECTIONAL FLUSHING PLAN NAS II

Listed is the sequence of hydrants to operate. Secure the noted hydrant before proceeding to the next location. Flow each hydrant the flowing water clears. Record the approximate minutes of flushing performed at each location.

Sequence	Hyd ID	Location
1	FH99B	(*)
2	FH100B	
3	FH113B	
4	FH109B	
5	FH106B	
6	FH107B	

Valves to be reopened:

Valve	Size	Location
	(mm)	
W36B	355	(*)
W15B	225	
W358B	200	
W20B	225	
W1B	355	

Note: If you proceed directly to Loop 9, Valves 1B, 15B, and 358B can remain closed.

Loop 9: (*)

Valves to Be Secured:

Valve	Size	Location
	(mm)	
W15B	225	(*)
W358B	200	
W4B	280	
W1B	355	
W2B	355	
W140B	200	

ATTACHMENT -1606001-07
UNIDIRECTIONAL FLUSHING PLAN NAS II

Listed is the sequence of hydrants to operate. Secure the noted hydrant before proceeding to the next location. Flow each hydrant the flowing water clears. Record the approximate minutes of flushing performed at each location.

Sequence	Hyd ID	Location
1	FH111B	(*)
2	FH119B	
3	FH114B	
4	FH115B	

Valves to be reopened:

Valve	Size	Location
	(mm)	
W15B	225	(*)
W358B	200	
W4B	280	
W1B	355	
W2B	355	
W140B	200	

Note: If you proceed directly to Loop 9, W140B can remain secured.

Loop 10: (*)

Valves to Be Secured:

Valve	Size	Location
	(mm)	
W140B	200	(*)
W9B	250	
W7B	225	

Listed is the sequence of hydrants to operate. Secure the noted hydrant before proceeding to the next location. Flow each hydrant the flowing water clears. Record the approximate minutes of flushing performed at each location.

Sequence	Hyd ID	Location
1	FH121B	(*)
2	FH110B	
3	FH126B	
4	FH128B	
5	FH129B	
6	FH122B	

ATTACHMENT -1606001-07
UNIDIRECTIONAL FLUSHING PLAN NAS II

Valves to be reopened:

Valve	Size	Location
	(mm)	
W140B	200	(*)
W9B	225	
W7B	225	

Loop 11: (*)

Valves to Be Secured:

Valve	Size	Location
	(mm)	
W38B	225	(*)

Listed is the sequence of hydrants to operate. Secure the noted hydrant before proceeding to the next location. Flow each hydrant the flowing water clears. Record the approximate minutes of flushing performed at each location.

Sequence	Hyd ID	Location
1	FH130B	(*)
2	FH137B	

Valves to be reopened:

Valve	Size	Location
	(mm)	
W38B	225	(*)

